Mind the Long Tail: Understanding the Difficulty of Delay Detection in Business Processes (Supplementary Appendix)

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Abstract. This supplementary appendix provides additional methodological, experimental, and implementation details that complement the main paper. It expands on the experimental setup by documenting the hyperparameter tuning procedures, including the search spaces and optimization strategies used across all evaluated methods. Furthermore, it presents detailed descriptions of the data-level and algorithm-level approaches for imbalanced regression, including formal definitions, implementation adaptations, and extended discussions of their practical considerations.

The appendix also includes comprehensive empirical results that are not fully reported in the main paper, such as per-dataset performance tables, and additional analyses of feature and label smoothing techniques. These results offer a more granular view of imbalanced regression and uncertainty-aware approaches across different target regions and datasets. Overall, this document aims to improve transparency, reproducibility, and interpretability of the experimental findings by providing the technical depth necessary to replicate and extend the work.

1 Introduction

This appendix supplements the main paper "Mind the Long Tail: Understanding the Difficulty of Delay Detection in Business Processes" by providing detailed methodological descriptions, implementation specifics, and extended experimental results. While the main paper focuses on the conceptual insights and high-level empirical findings, this document offers the technical depth required for full reproducibility and a more comprehensive understanding of the evaluated approaches.

First, we describe the hyperparameter tuning procedure used throughout the study, including the Bayesian optimization framework, search spaces, and

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method-specific configurations. This is intended to clarify how model configurations were selected and to ensure that the reported results can be reliably reproduced.

Second, we provide a detailed account of the data-level and algorithm-level approaches for imbalanced regression. This includes formal definitions of the methods, implementation details tailored to the predictive process monitoring setting, and clarifications on how these techniques interact with sequential event data.

Third, we extend the analysis of predictive uncertainty presented in the main paper by incorporating additional uncertainty estimation techniques. In particular, we examine heteroscedastic uncertainty using both survival-based modeling and quantile regression, providing a more comprehensive view of how uncertainty relates to remaining time across different modeling paradigms.

Finally, the appendix presents extended experimental results that complement those in the main paper. These include per-dataset performance tables, additional comparisons across methods, and further analyses of delay detection and uncertainty-aware modeling.

Together, these additions provide a comprehensive and transparent view of the methodological choices and empirical behavior underlying the results reported in the main paper.

2 Analysis of Heteroscedastic Uncertainty

we extend the analysis presented in the main paper by incorporating an additional uncertainty-aware approach based on quantile regression. Specifically, alongside the previously reported Spearman correlations between the ground-truth remaining time y and (i) the absolute prediction error of the standard LSTM model and (ii) the prediction interval width of the survival-based model, we further examine the correlation between y and the prediction interval width obtained from a quantile regression model. This additional analysis allows us to assess whether the relationship between predictive uncertainty and remaining time is consistent across different uncertainty estimation techniques, thereby providing a more comprehensive evaluation of the observed effects. We first breifly inroduce quantile regression and then present the results.

Quantile regression. Quantile regression directly estimates conditional quantiles of the remaining time distribution. Instead of predicting a single value, the model outputs multiple quantiles $q_\tau(x)$ for $\tau \in \{0.1, 0.5, 0.6, 0.9, 0.95, 0.99\}$ using a shared LSTM encoder with separate output heads. The model is trained with the pinball loss, which minimizes asymmetric absolute errors for each quantile. Prediction intervals can then be obtained from the estimated quantiles; in our analysis we use the width of the central 80% interval $q_{0.9}(x) - q_{0.1}(x)$ as a measure of predictive dispersion.

Anlysis. Table 1 extends the previous analysis by including prediction interval widths obtained from quantile regression. Overall, the results reinforce the

Table 1: Spearman correlation between ground-truth remaining time y and (i) absolute prediction error $|e|$ and (ii) prediction interval (PI) width produced by quantile regression (QR) and the survival model (Surv).

Dataset	$\rho(y, \|e\|)$	$\rho(y, \mathrm{PI}_{QR})$	$\rho(y, \mathrm{PI}_{Surv})$	Dataset	$\rho(y, \|e\|)$	$\rho(y, \mathrm{PI}_{QR})$	$\rho(y, \mathrm{PI}_{Surv})$
P2P	0.89	0.63	0.46	Helpdesk	−0.41	−0.40	−0.53
BPIC17W	0.38	0.35	−0.43	Sepsis	0.51	0.27	0.39
BPIC15-1	0.30	0.55	0.55	BPIC20ID	0.64	0.81	0.80
BPIC15-2	−0.20	0.31	−0.22	BPIC20DD	0.47	0.48	0.21
BPIC15-3	0.68	0.63	0.64	BPIC20PTC	0.56	0.70	0.67
BPIC15-4	−0.69	0.39	0.09	BPIC20TPD	0.61	0.69	0.68
BPIC15-5	0.60	0.44	0.32	BPIC20RFP	0.55	0.48	0.22

presence of heteroscedasticity. In the majority of event logs, both uncertainty estimates—quantile regression and survival-based—exhibit a positive correlation with the true remaining time, broadly consistent with the behavior observed for the absolute prediction error. In several datasets (e.g., P2P, BPIC20ID, BPIC20PTC, BPIC20TPD), both uncertainty measures show moderate to strong positive correlations, indicating that the increase in uncertainty for long-running cases is robust across modeling approaches.

At the same time, the extended results highlight some differences between the two uncertainty estimation techniques. While the two measures are often aligned, discrepancies arise in certain logs. For instance, BPIC17W shows a positive correlation for quantile regression but a negative one for the survival model, and BPIC15-4 exhibits negative error correlation but positive correlations for both uncertainty estimates. Similarly, BPIC15-2 displays mixed behavior, with quantile regression indicating a positive association while the other measures remain negative.

These variations suggest that, although the overall trend of increasing uncertainty with remaining time persists, the strength and even direction of this relationship can depend on the chosen uncertainty estimation method. Nevertheless, the dominant pattern remains consistent: longer-running cases tend to be associated with higher predictive uncertainty, confirming that heteroscedasticity is not an artifact of a specific model, but a consistent property of the data across different uncertainty estimation techniques.

These findings motivate the use of uncertainty-aware modeling for downstream tasks such as delay detection, which we analyze in detail in the subsequent sections.

3 Hyperparameter Tuning

We tune hyperparameters using Bayesian optimization implemented via Ax (Add reference to the library). Each configuration is evaluated on the validation set, and the objective is to minimize validation loss. The optimization starts with Sobol sampling followed by Bayesian optimization. The search space is defined per method. Common parameters include the learning rate (log-scale) and loss-

Table 2: Hyperparameter search space for each method. Fixed values indicate no tuning.

Approach	Hyperparameter	Search space
All	Learning rate	$[10^{-5}, 10^{-2}]$ (log)
Vanilla	Loss	MAE (fixed)
SMOGN	k	(default = 5)
	Relevance threshold	(default = 0.5)
	Over-sampling ratio	(default = 10)
CSW	Loss	{MAE, MSE, Huber}
	Reweighting	{inverse, sqrt_inv}
	Bins	{10, 20, 50}
EAL	Loss	{Focal-MAE, Focal-MSE}
	β	$[10^{-2}, 1]$ (log)
	γ	$[0.5, 5]$
BMSE	Loss	BMSE (fixed)
SERA	Extreme type	{high, both}
	Asymmetry	{False, True}
Survival	Bins	{10, 15, 20}
	Binning	{quantile, hybrid_tail}
	Tail fraction	{0.1, 0.2, 0.3}
	Tail bin fraction	{0.3, 0.4, 0.5}
	Prediction type	{mean, median}

specific settings, while method-specific parameters are activated only when relevant. The number of trials is 10 for Vanilla, and 45 for all other approaches. Table 2 summarizes the tuned hyperparameters and their search spaces.

4 Data-level Approaches

This section provides additional details on SMOGN as a data-level approach for handling imbalanced regression. We report the specific configurations and hyperparameters employed in our study, as summarized in Table 3.

5 Algorithm-level Approaches

Algorithm-level approaches mitigate imbalance by modifying the regression objective to assign greater weight to underrepresented target regions. They can be combined with feature and label distribution smoothing techniques as we explain in this section. We first evaluate the performance of algorithm-level approaches and then evaluate their combination with feature and label smoothing.

Table 3: Performance difference between the LSTM trained with SMOGN and the same model trained on the original (imbalanced) data (Δ = SMOGN − Original). Negative values indicate an improvement (lower nMAE).

Log	All	Many	Med.	Few
P2P	-0.06	0.03	0.00	-0.18
BPIC17W	-0.01	0.05	-0.06	-0.16
BPIC15-1	0.16	0.17	0.22	-0.39
BPIC15-2	1.34	1.33	1.51	-1.85
BPIC15-3	0.46	0.42	0.87	-1.20
BPIC15-4	0.05	0.05	0.02	-0.01
BPIC15-5	0.07	0.04	0.14	0.20
Helpdesk	-0.01	0.01	-0.08	-0.11
Sepsis	0.05	0.01	0.13	0.14
BPIC20ID	0.03	0.04	0.00	-0.06
BPIC20DD	0.00	0.07	-0.05	-0.16
BPIC20PTC	0.04	0.08	-0.02	-0.33
BPIC20TPD	0.05	0.13	-0.07	-0.05
BPIC20RFP	0.01	0.08	-0.07	-0.07
Average	0.16 (+17.5%)	0.18 (+31.4%)	0.18 (+20.5%)	-0.30 (-8.0%)

5.1 Reweighting the Loss Function

We consider cost-sensitive weighting, distribution-balanced objectives, error-aware losses, and relevance-based error measures.

Cost-Sensitive Weighting (CSW). The empirical density of $p_{\text{train}}(y)$ can be used to assign static loss weights to different target ranges. In practice, this is done by discretizing the target domain into I consecutive bins of equal width and mapping each target value to a bin index $i \in \{1, \ldots, I\}$. Let n_i denote the number of training samples in bin i and N the total number of samples. The empirical bin frequency is $\hat{p}_i = n_i/N$. The loss weight in Eq. 2 for samples in bin i is then set to

$$w_i = \frac{1}{\hat{p}_i} \quad \text{or} \quad w_i = \frac{1}{\sqrt{\hat{p}_i}} \tag{1}$$

thereby down-weighting overrepresented regions and up-weighting underrepresented ones [6].

$$\mathcal{L}(\theta, \psi) = \begin{cases} \frac{1}{N}\sum_{i=1}^{N} w_i\,|\hat{y}_i - y_i|, & \text{if absolute error is used,} \\ \frac{1}{N}\sum_{i=1}^{N} w_i\,(\hat{y}_i - y_i)^2, & \text{if squared error is used.} \end{cases} \tag{2}$$

Balanced Mean Squared Error (BMSE). BMSE extends static bin-based weighting by incorporating the training target distribution $p_{\text{train}}(y)$ directly into the regression objective. We adopt the batch-based Monte Carlo (BMC) variant of BMSE [2], which treats the target values in a mini-batch $B_y = \{y_{(1)}, \ldots, y_{(n)}\}$ as samples from $p_{\text{train}}(y)$ and uses them to normalize prediction errors within the batch. For a prediction $\hat{y}$ and target y, the BMC loss is defined as:

$$\mathcal{L}_{\text{BMC}} = -\log \frac{\exp\left(-\frac{\|\hat{y}-y\|^2}{2\sigma^2_{\text{noise}}}\right)}{\sum_{y' \in B_y} \exp\left(-\frac{\|\hat{y}-y'\|^2}{2\sigma^2_{\text{noise}}}\right)} \tag{3}$$

This objective can be interpreted as a softmax over squared distances within the batch: the numerator favors small error for the true target, while the denominator contrasts it against errors to other batch targets. As a result, prediction errors in densely populated target regions contribute less relative weight than errors in sparse regions. The temperature parameter $\tau = 2\sigma^2_{\text{noise}}$ controls the sharpness of this contrast. Following [2], we treat σ_{noise} as a learnable parameter and optimize it jointly with the model parameters during training.

Error-Aware Loss (EAL). Error-aware loss functions adapt sample weights based on the magnitude of the model's prediction error. For instance, Focal-R [6] loss emphasizes hard examples by down-weighting those with small errors through a sigmoid-based scaling factor. It is defined as

$$\mathcal{L}_{\text{Focal-R}} = \frac{1}{N} \sum_{i=1}^{N} \sigma(|\beta e_i|)^{\gamma} \, e_i, \tag{4}$$

where e_i denotes the absolute or squared error of the i-th sample, $\sigma(\cdot)$ is the sigmoid function, and β and γ are hyperparameters controlling the scaling and focusing strength. Larger errors therefore receive relatively higher weights, increasing the contribution of difficult examples during optimization.

Squared Error Relevance Area (SERA). SERA is a relevance-based criterion for imbalanced regression that can be used both as a training objective and as an evaluation metric [4,3]. It relies on a relevance function $\phi : Y \to [0, 1]$ that assigns each target value an importance score, enabling non-uniform preference over the continuous target domain [5,1]. In this work, ϕ is automatically derived from the training target distribution using box-plot statistics, as proposed by Ribeiro and Moniz [3]. This yields higher relevance scores for extreme remaining time values.

Given a relevance threshold r, the set of relevant samples is defined as $\mathcal{D}^r = \{(\mathbf{x}, y) \in \mathcal{D} : \phi(y) \geq r\}$. SERA aggregates squared errors over relevance-defined subsets across all thresholds:

$$\mathcal{L}_{\text{SERA}} = \int_0^1 \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}^r} (y_i - \hat{y}_i)^2 \, dr. \tag{5}$$

Errors on high-relevance targets are counted across a wider range of thresholds and therefore receive greater overall weight, while low-relevance targets contribute less. This mechanism emphasizes extreme values without fully discarding the rest of the target range [4].

Evaluation. We now evaluate the effectiveness of the considered algorithm-level approaches across all 14 event logs. The results are reported in Table 4, Table 5, and Table 6. We report normalized MAE (nMAE) both overall and across target frequency regions (Many, Medium, Few), enabling a fine-grained analysis of how different methods handle imbalance. This breakdown is particularly useful for assessing performance on underrepresented regions (Few), where imbalance effects are most pronounced. We compare standard training (Vanilla) with both data-level (SMOGN) and algorithm-level approaches to highlight their relative strengths and trade-offs across datasets.

Statistical Analysis of Algorithm-Level Approaches. To assess whether the observed performance differences between algorithm-level imbalanced regression approaches are statistically significant, we employ a two-stage non-parametric testing procedure. First, we apply the *Friedman test*, which evaluates whether multiple approaches exhibit equivalent performance across multiple event logs.

When the Friedman test indicates significant differences, we perform a *Nemenyi post-hoc* test to identify which pairs of methods differ significantly. This test compares the average ranks of all pairs of approaches and accounts for multiple comparisons. In addition, for consistency with the main paper, we report significant pairwise differences using the *Wilcoxon signed-rank test*, while the Nemenyi test provides a complementary global view of pairwise differences across all methods. All tests are conducted separately for each target region (Many, Medium, Few) as well as overall (All), using nMAE as the evaluation metric.

Table 7 summarizes the Friedman statistics and average ranks, while Table 8, Table 9, Table 10, and Table 11 report detailed pairwise Nemenyi p-values.

Many-shot Region. In the many-shot region, the Friedman test indicates significant differences across methods ($\chi^2 = 43.26$, $p < 0.001$). The average ranks confirm that Vanilla performs best, followed by CSW and EAL, while SERA ranks last. The Nemenyi post-hoc results (Table 8) show that SERA is significantly worse than Vanilla ($p < 0.001$), CSW ($p = 0.006$), and EAL ($p = 0.015$). No other pairwise differences reach statistical significance, indicating that the top-performing methods (Vanilla, CSW, EAL) behave similarly in dense regions of the target space: when sufficient data is available, simple objectives such as MAE remain competitive, and more complex imbalance-aware objectives do not yield additional benefits.

Medium-shot Region. For the medium-shot region, the Friedman test again reveals significant differences ($\chi^2 = 27.43$, $p < 0.001$). The ranking shows a more balanced picture, with CSW and BMSE achieving the best average ranks, followed by EAL and Vanilla, while SERA again performs worst. The Nemenyi results (Table 9) indicate that SERA is significantly worse than CSW ($p = 0.012$) and BMSE ($p = 0.020$). Differences among the remaining methods are not statistically significant, suggesting that performance in moderately populated regions is relatively similar across most approaches. Overall, these results reinforce that SERA underperforms outside the extreme regions, while methods such as CSW and BMSE offer slight advantages without clear dominance.

Table 4: nMAE of different imbalanced regression approaches across five BPIC15 event logs. Results are reported overall (All) and across target frequency groups (Many, Medium, Few). Lower values indicate better performance.

Log	IR	nMAE ↓ All	nMAE ↓ Many	nMAE ↓ Med.	nMAE ↓ Few
BPIC15-1	Vanilla	0.99 (0.07)	0.88 (0.09)	0.87 (0.09)	4.90 (0.35)
	SMOGN	1.15 (0.09)	1.05 (0.09)	1.09 (0.12)	4.51 (0.15)
	CSW	0.89 (0.01)	0.76 (0.04)	**0.82 (0.05)**	5.02 (0.14)
	EAL	**0.87 (0.04)**	**0.70 (0.07)**	0.88 (0.07)	5.03 (0.07)
	BMSE	1.33 (0.13)	1.34 (0.18)	1.08 (0.18)	4.10 (0.12)
	SERA	5.52 (0.62)	6.12 (0.78)	4.63 (0.39)	**1.51 (0.40)**
	Survival	1.69 (0.35)	1.54 (0.43)	1.83 (0.25)	3.77 (0.28)
BPIC15-2	Vanilla	1.12 (0.12)	1.14 (0.15)	**0.93 (0.11)**	4.04 (0.33)
	SMOGN	2.46 (0.44)	2.47 (0.53)	2.44 (0.10)	2.19 (0.23)
	CSW	**1.11 (0.19)**	**1.06 (0.24)**	1.27 (0.17)	3.32 (0.47)
	EAL	1.25 (0.19)	1.30 (0.21)	0.98 (0.23)	3.83 (0.27)
	BMSE	1.36 (0.35)	1.45 (0.43)	**0.93 (0.14)**	3.80 (0.51)
	SERA	6.43 (0.00)	6.91 (0.00)	4.57 (0.00)	**1.22 (0.00)**
	Survival	2.94 (0.13)	3.07 (0.17)	2.42 (0.10)	2.40 (0.26)
BPIC15-3	Vanilla	0.88 (0.03)	0.41 (0.06)	1.06 (0.01)	5.11 (0.13)
	SMOGN	1.34 (0.11)	0.83 (0.14)	1.93 (0.14)	3.91 (0.11)
	CSW	**0.86 (0.06)**	**0.39 (0.05)**	**1.03 (0.06)**	4.92 (0.35)
	EAL	0.90 (0.03)	0.42 (0.04)	1.11 (0.02)	4.95 (0.09)
	BMSE	1.04 (0.09)	0.61 (0.10)	1.26 (0.13)	4.57 (0.21)
	SERA	4.50 (0.13)	5.00 (0.18)	3.94 (0.08)	**1.95 (0.10)**
	Survival	1.35 (0.08)	0.85 (0.07)	2.09 (0.13)	3.14 (0.14)
BPIC15-4	Vanilla	1.09 (0.02)	1.12 (0.03)	0.45 (0.03)	1.66 (0.04)
	SMOGN	1.14 (0.09)	1.17 (0.11)	0.47 (0.12)	1.65 (0.13)
	CSW	1.09 (0.03)	1.12 (0.03)	0.42 (0.03)	1.64 (0.05)
	EAL	**1.08 (0.04)**	**1.10 (0.04)**	0.45 (0.03)	1.67 (0.06)
	BMSE	1.26 (0.03)	1.31 (0.04)	**0.30 (0.03)**	1.38 (0.08)
	SERA	2.84 (0.00)	2.95 (0.00)	1.31 (0.00)	**0.26 (0.00)**
	Survival	1.67 (0.08)	1.73 (0.09)	0.82 (0.08)	0.92 (0.12)
BPIC15-5	Vanilla	**0.81 (0.03)**	**0.51 (0.05)**	1.11 (0.06)	3.68 (0.14)
	SMOGN	0.88 (0.03)	0.55 (0.03)	1.25 (0.18)	3.88 (0.20)
	CSW	0.83 (0.02)	0.56 (0.06)	1.08 (0.16)	3.61 (0.11)
	EAL	0.83 (0.01)	0.56 (0.06)	1.05 (0.13)	3.48 (0.10)
	BMSE	1.06 (0.03)	1.04 (0.06)	**0.61 (0.05)**	3.25 (0.06)
	SERA	6.08 (0.11)	6.70 (0.12)	4.87 (0.12)	**2.51 (0.19)**
	Survival	1.18 (0.11)	1.03 (0.12)	1.37 (0.28)	2.57 (0.25)

Table 5: nMAE of different imbalanced regression approaches across five BPIC20 event logs. Results are reported overall (All) and across target frequency groups (Many, Medium, Few). Lower values indicate better performance.

Log	IR	nMAE ↓			
		All	Many	Med.	Few
BPIC20ID	Vanilla	**0.72 (0.01)**	**0.44 (0.00)**	1.13 (0.01)	5.05 (0.03)
	SMOGN	0.75 (0.01)	0.48 (0.00)	1.13 (0.02)	4.99 (0.03)
	CSW	**0.72 (0.01)**	**0.44 (0.00)**	1.12 (0.02)	5.07 (0.04)
	EAL	**0.72 (0.00)**	0.45 (0.00)	**1.11 (0.00)**	5.06 (0.02)
	BMSE	0.83 (0.01)	0.58 (0.03)	1.19 (0.02)	4.67 (0.02)
	SERA	5.24 (0.02)	6.01 (0.02)	3.82 (0.02)	**2.16 (0.00)**
	Survival	1.32 (0.04)	0.79 (0.02)	2.28 (0.10)	3.61 (0.12)
BPIC20DD	Vanilla	**0.87 (0.00)**	**0.38 (0.01)**	0.62 (0.01)	3.34 (0.02)
	SMOGN	**0.87 (0.00)**	0.45 (0.01)	0.57 (0.01)	3.18 (0.03)
	CSW	**0.87 (0.00)**	0.40 (0.01)	0.60 (0.01)	3.30 (0.02)
	EAL	**0.87 (0.00)**	0.42 (0.01)	0.58 (0.01)	3.25 (0.02)
	BMSE	0.94 (0.01)	0.68 (0.02)	**0.56 (0.00)**	2.72 (0.02)
	SERA	2.71 (0.03)	3.35 (0.04)	2.10 (0.04)	**1.43 (0.01)**
	Survival	1.42 (0.07)	1.26 (0.06)	1.48 (0.14)	1.94 (0.03)
BPIC20PTC	Vanilla	**0.79 (0.01)**	**0.48 (0.01)**	**1.08 (0.01)**	4.59 (0.09)
	SMOGN	0.83 (0.00)	0.56 (0.01)	**1.06 (0.01)**	4.26 (0.04)
	CSW	0.98 (0.01)	0.73 (0.01)	1.28 (0.03)	3.37 (0.06)
	EAL	0.83 (0.00)	0.55 (0.01)	1.09 (0.01)	4.26 (0.02)
	BMSE	1.00 (0.01)	0.74 (0.01)	1.30 (0.02)	3.31 (0.05)
	SERA	2.92 (0.10)	2.64 (0.13)	3.65 (0.07)	**0.98 (0.01)**
	Survival	1.71 (0.04)	1.12 (0.04)	2.95 (0.04)	1.08 (0.04)
BPIC20TPD	Vanilla	**0.76 (0.01)**	**0.49 (0.02)**	0.86 (0.01)	2.99 (0.09)
	SMOGN	0.81 (0.01)	0.62 (0.01)	**0.79 (0.02)**	2.94 (0.06)
	CSW	**0.76 (0.01)**	0.50 (0.01)	0.85 (0.02)	2.96 (0.06)
	EAL	**0.76 (0.00)**	0.50 (0.01)	0.85 (0.01)	2.96 (0.04)
	BMSE	0.88 (0.02)	0.71 (0.02)	0.89 (0.04)	2.53 (0.08)
	SERA	3.34 (0.01)	4.09 (0.01)	2.30 (0.01)	**1.06 (0.00)**
	Survival	1.16 (0.03)	1.03 (0.04)	1.34 (0.03)	1.58 (0.06)
BPIC20RFP	Vanilla	**0.87 (0.00)**	**0.36 (0.01)**	0.64 (0.01)	2.99 (0.01)
	SMOGN	0.88 (0.00)	0.44 (0.01)	0.57 (0.01)	2.92 (0.02)
	CSW	**0.87 (0.00)**	0.38 (0.01)	0.62 (0.01)	2.96 (0.02)
	EAL	**0.87 (0.00)**	0.42 (0.01)	0.59 (0.01)	2.85 (0.01)
	BMSE	0.92 (0.02)	0.65 (0.08)	**0.56 (0.01)**	2.45 (0.13)
	SERA	2.91 (0.03)	3.59 (0.03)	2.56 (0.04)	**1.22 (0.01)**
	Survival	1.32 (0.03)	1.23 (0.03)	1.47 (0.05)	1.38 (0.02)

Table 6: nMAE of different imbalanced regression approaches across other four event logs. Results are reported overall (All) and across target frequency groups (Many, Medium, Few). Lower values indicate better performance.

Log	IR	nMAE ↓ All	Many	Med.	Few
P2P	Vanilla	0.88 (0.01)	**0.14 (0.01)**	0.21 (0.00)	1.83 (0.02)
	SMOGN	**0.82 (0.02)**	0.17 (0.01)	0.21 (0.02)	1.65 (0.05)
	CSW	0.88 (0.07)	0.29 (0.07)	**0.18 (0.05)**	1.70 (0.13)
	EAL	0.83 (0.01)	0.24 (0.04)	0.26 (0.06)	1.59 (0.06)
	BMSE	0.84 (0.03)	0.30 (0.05)	0.33 (0.08)	1.53 (0.05)
	SERA	0.93 (0.12)	0.94 (0.32)	0.59 (0.13)	**1.07 (0.05)**
	Survival	0.84 (0.03)	0.47 (0.09)	0.51 (0.13)	1.31 (0.08)
BPIC17W	Vanilla	0.89 (0.00)	**0.50 (0.01)**	0.78 (0.01)	3.75 (0.02)
	SMOGN	**0.88 (0.00)**	0.55 (0.02)	0.72 (0.02)	3.59 (0.02)
	CSW	0.89 (0.01)	0.51 (0.01)	0.77 (0.02)	3.75 (0.03)
	EAL	0.89 (0.00)	**0.50 (0.01)**	0.78 (0.01)	3.75 (0.02)
	BMSE	1.01 (0.02)	0.87 (0.04)	**0.58 (0.01)**	3.28 (0.04)
	SERA	3.88 (0.13)	4.68 (0.17)	3.04 (0.09)	**1.20 (0.03)**
	Survival	1.54 (0.01)	1.52 (0.01)	1.44 (0.02)	1.96 (0.01)
Helpdesk	Vanilla	0.87 (0.01)	**0.87 (0.01)**	0.62 (0.01)	1.44 (0.01)
	SMOGN	**0.86 (0.02)**	0.88 (0.03)	0.54 (0.04)	1.33 (0.03)
	CSW	0.91 (0.01)	0.97 (0.01)	**0.46 (0.01)**	1.25 (0.01)
	EAL	0.91 (0.01)	0.96 (0.01)	0.47 (0.01)	1.27 (0.01)
	BMSE	0.91 (0.02)	0.96 (0.03)	0.47 (0.02)	1.27 (0.02)
	SERA	1.82 (0.02)	2.17 (0.02)	0.61 (0.01)	**0.31 (0.00)**
	Survival	0.88 (0.02)	0.93 (0.03)	0.48 (0.02)	1.27 (0.03)
Sepsis	Vanilla	**0.94 (0.00)**	0.27 (0.02)	2.03 (0.03)	7.76 (0.04)
	SMOGN	0.99 (0.00)	0.28 (0.00)	2.16 (0.01)	7.90 (0.01)
	CSW	**0.94 (0.00)**	**0.23 (0.01)**	2.14 (0.03)	7.86 (0.02)
	EAL	**0.94 (0.00)**	0.23 (0.01)	2.12 (0.03)	7.85 (0.02)
	BMSE	1.71 (0.10)	1.62 (0.14)	**1.32 (0.02)**	6.36 (0.12)
	SERA	6.91 (0.02)	7.70 (0.02)	5.34 (0.02)	**0.88 (0.01)**
	Survival	2.21 (0.33)	2.18 (0.41)	1.99 (0.27)	4.58 (0.36)

Table 7: Friedman test and Nemenyi post-hoc summary (nMAE)

Mode	χ^2	p	Avg. ranks	Significant pairs
Many	43.26	< 0.001	Vanilla (1.64), CSW (2.14), EAL (2.36), BMSE (3.86), SERA (5.00)	Vanilla vs SERA (0.001); CSW vs SERA (0.006); EAL vs SERA (0.015)
Medium	27.43	< 0.001	CSW (2.21), BMSE (2.36), EAL (2.64), Vanilla (2.86), SERA (4.93)	CSW vs SERA (0.012); BMSE vs SERA (0.020)
Few	44.40	< 0.001	SERA (1.00), BMSE (2.14), CSW (3.57), EAL (3.79), Vanilla (4.50)	Vanilla vs BMSE (0.042); Vanilla vs SERA (0.000); CSW vs SERA (0.020); EAL vs SERA (0.009)
All	39.60	< 0.001	EAL (1.93), CSW (2.00), Vanilla (2.36), BMSE (3.71), SERA (5.00)	Vanilla vs SERA (0.015); CSW vs SERA (0.004); EAL vs SERA (0.003)

Table 8: Pairwise Nemenyi post-hoc p-values for the Vanilla model trained with MAE and the algorithm-level imbalanced regression approaches (across **Many** target regions).

Approach	Vanilla	CSW	EAL	BMSE	SERA
Vanilla	1.000	0.976	0.916	0.067	< 0.001
CSW	0.976	1.000	0.999	0.252	0.006
EAL	0.916	0.999	1.000	0.388	0.015
BMSE	0.067	0.252	0.388	1.000	0.658
SERA	< 0.001	0.006	0.015	0.658	1.000

Few-shot Region. A different pattern emerges in the few-shot region. The Friedman test confirms strong differences ($\chi^2 = 44.40$, $p < 0.001$), and the ranking clearly shows SERA as the best-performing method (rank 1.00), followed by BMSE. The Nemenyi analysis (Table 10) provides strong evidence for this improvement:

– SERA significantly outperforms Vanilla ($p < 0.001$), CSW ($p = 0.020$), and EAL ($p = 0.009$)
– BMSE also significantly outperforms Vanilla ($p = 0.042$)

No significant differences are observed among CSW, EAL, and BMSE beyond this. These findings confirm that relevance-based objectives such as SERA are particularly effective in sparse regions, where emphasizing rare targets is beneficial.

Overall Performance (All Regions). When considering all target regions jointly, the Friedman test again indicates significant differences ($\chi^2 = 39.60$, $p < 0.001$). The average ranks show that EAL and CSW achieve the best overall performance, followed by Vanilla, while BMSE and especially SERA perform worse. The Nemenyi results (Table 11) confirm that SERA is significantly worse than

Table 9: Pairwise Nemenyi post-hoc p-values for the Vanilla model trained with MAE and the algorithm-level imbalanced regression approaches (across **Medium** target regions).

Approach	Vanilla	CSW	EAL	BMSE	SERA
Vanilla	1.000	0.942	0.999	0.976	0.102
CSW	0.942	1.000	0.987	1.000	0.012
EAL	0.999	0.987	1.000	0.997	0.053
BMSE	0.976	1.000	0.997	1.000	0.020
SERA	0.102	0.012	0.053	0.020	1.000

Table 10: Pairwise Nemenyi post-hoc p-values for the Vanilla model trained with MAE and the algorithm-level imbalanced regression approaches (across **Few** target region).

Approach	Vanilla	CSW	EAL	BMSE	SERA
Vanilla	1.000	0.807	0.916	0.042	< 0.001
CSW	0.807	1.000	0.999	0.440	0.020
EAL	0.916	0.999	1.000	0.294	0.009
BMSE	0.042	0.440	0.294	1.000	0.658
SERA	< 0.001	0.020	0.009	0.658	1.000

Vanilla ($p = 0.015$), CSW ($p = 0.004$), and EAL ($p = 0.003$). No other pairwise differences are statistically significant, indicating that the top-performing approaches are again relatively close in performance. These results highlight an important trade-off: although SERA excels in the few-shot region, this advantage does not translate into improved overall performance. Instead, methods such as EAL and CSW provide a better balance across the entire target distribution.

5.2 Feature and Label Smoothing

Algorithm-level imbalanced regression approaches can also be combined with feature and label distribution smoothing techniques that exploit the continuity of the target space.

Label Distribution Smoothing. In imbalanced regression, simply counting how often each target value appears in the training data can be misleading. Unlike classification, target values are continuous and nearby values are typically similar. As a result, prediction difficulty does not necessarily follow the raw label frequency. Label Distribution Smoothing (LDS) addresses this issue by estimating an *effective* label density that also accounts for neighboring target values.

Table 11: Pairwise Nemenyi post-hoc p-values for the Vanilla model trained with MAE and the algorithm-level imbalanced regression approaches (across **All** target regions).

Approach	Vanilla	CSW	EAL	BMSE	SERA
Vanilla	1.000	0.993	0.987	0.494	0.015
CSW	0.993	1.000	1.000	0.252	0.004
EAL	0.987	1.000	1.000	0.214	0.003
BMSE	0.494	0.252	0.214	1.000	0.549
SERA	0.015	0.004	0.003	0.549	1.000

Concretely, the continuous target space is first discretized into bins, as in cost-sensitive weighting. Instead of directly using the bin counts, LDS smooths them by spreading each bin's mass to its neighbors using a symmetric kernel. This produces a smoothed label distribution in which nearby target values support each other. Bins located in sparse regions but surrounded by populated neighbors are therefore treated as less rare than their raw counts suggest. The resulting smoothed density can then replace the empirical distribution in standard loss re-weighting schemes, yielding more stable and more realistic weights [6].

Feature Distribution Smoothing (FDS). Under skewed target distributions, the encoder $\mathbf{z} = g(\mathbf{x}, \theta)$ tends to learn latent features that are dominated by well-represented target regions. FDS reduces this effect by smoothing feature statistics across nearby target values. After the encoder produces latent features, the target space is discretized into bins and the mean and covariance of latent features within each bin are estimated. FDS smooths these statistics across neighboring bins using a symmetric kernel, assuming that nearby target values should have similar latent representations. The resulting features $\tilde{\mathbf{z}}$ are then passed to the prediction head: $\hat{y} = h(\tilde{\mathbf{z}}, \psi)$. Implemented as a lightweight feature calibration layer before the prediction head, FDS can be integrated into any neural network, complementing previous algorithm-level approaches [6].

Evaluation. We next evaluate whether combining feature and label smoothing with imbalance-aware objectives improves prediction performance. Note that LDS can only be combined with approaches that rely on sample reweighting. Since SERA employs a relevance-weighted loss and BMSE uses a pairwise probabilistic objective that does not incorporate sample weights, LDS is not directly compatible with these approaches.

Table 12 reports the relative change in nMAE after combining algorithm-level imbalanced regression methods with feature and label smoothing. Overall, smoothing produces only minor and inconsistent effects, with most Δ values being positive, indicating increased prediction error compared to the corresponding baseline models. For CSW, EAL, and BMSE, feature distribution smoothing results in negligible changes and no statistically significant improvements. When combined with SERA, FDS yields significant improvements in the overall and many-shot regions (-7.97% and -9.05%, p = 0.011), but does not improve performance in the few-shot region. Label distribution smoothing exhibits unstable behavior: while it slightly improves the few-shot region for EAL (-3.31%, $p = 0.013$), it substantially degrades performance in the many-shot region (+11.71%, p = 0.035). Overall, smoothing techniques frequently increase prediction errors and do not consistently improve performance, particularly in the few-shot region corresponding to extreme remaining times. Statistical significance is assessed using a paired Wilcoxon signed-rank test across datasets, where p-values below 0.05 indicate a consistent difference between smoothed and baseline models. These findings further support the observation that the challenge of predicting extreme remaining times cannot be addressed solely through imbalance-aware objectives or distribution smoothing techniques.

Table 12: Average relative change in nMAE (%) after applying smoothing to imbalanced regression approaches. Negative values indicate improvement.Statistical significance is assessed with a paired Wilcoxon signed-rank test across datasets.

IR	Smoothing	All		Many		Medium		Few	
		Δ	p	Δ	p	Δ	p	Δ	p
CSW	FDS	-0.14%	0.715	-1.11%	0.426	+5.96%	0.296	+3.45%	0.761
EAL	FDS	-0.55%	1.000	-1.99%	0.296	+5.49%	0.217	+0.88%	0.715
BMSE	FDS	-0.95%	0.091	-1.64%	0.296	+7.11%	0.808	+2.88%	0.391
SERA	FDS	**-7.97%**	**0.011**	**-9.05%**	**0.011**	-1.01%	0.194	+0.70%	0.626
CSW	LDS	+0.11%	0.391	+0.85%	0.502	+4.73%	0.761	+3.26%	0.670
EAL	LDS	+2.85%	0.078	**+11.71%**	**0.035**	+8.36%	0.391	**-3.31%**	**0.013**
CSW	LDS+FDS	-1.55%	0.391	-2.87%	0.326	+7.64%	0.358	+6.43%	0.326
EAL	LDS+FDS	+2.32%	0.761	+4.25%	0.670	+12.86%	0.135	+0.90%	0.903

6 Delay Detection via Uncertainty Modeling

This section provides additional analysis and detailed results for the delay detection task introduced in the main paper. While the main paper focuses on the overall performance improvements achieved by incorporating predictive uncertainty, here we present complementary evidence that offers deeper insight into the behavior of both the baseline and uncertainty-aware models.

In particular, we analyze classification outcomes through confusion matrices, examine how early delayed cases can be detected during process execution, and evaluate the robustness of the proposed approach under a stricter delay definition. These analyses provide a more fine-grained understanding of where the performance gains originate and how uncertainty contributes to improved delay detection.

Sparsification error. To evaluate whether the predicted uncertainty correctly identifies unreliable predictions, we perform a sparsification analysis. Sparsification analysis evaluates whether uncertainty estimates correctly identify unreliable predictions by ranking test instances according to predicted uncertainty and progressively removing the most uncertain ones (1% at a time), while computing the mean absolute error of the remaining predictions. If uncertainty estimates are informative, removing highly uncertain predictions should monotonically reduce the remaining prediction error. For reference, we also compute two baselines: an oracle ranking that removes predictions according to their true error, and a random ranking that removes instances uniformly at random. The oracle represents the optimal ranking, while the random curve represents an uninformative uncertainty estimate. Because the evaluated datasets have very different time scales, sparsification analysis is performed using normalized errors. Following the normalization used for nMAE, errors are divided by the mean absolute deviation of the target values from the median of the training set. This normalization preserves the ranking of predictions while enabling direct comparison across datasets.

Table 13 summarizes the normalized sparsification metrics. The normalized Area Under the Sparsification Error curve (nAUSE) measures the deviation from the oracle ranking, while the normalized Area Under the Random Gain curve (nAURG) measures the improvement over random removal. Lower nAUSE and higher nAURG indicate better uncertainty ranking. The ratio ρ_{SP} is obtained by deviding nAURG by the sum of nAURG and nAUSE, and hence, summarizes how close the uncertainty ranking is to the oracle relative to random.

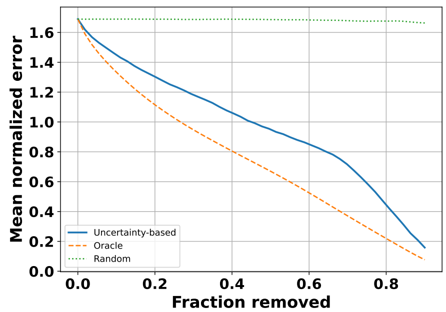
Overall, uncertainty-based ranking consistently improves over random removal in most event logs. In 7 out of 14 logs (BPIC15-1, BPIC15-3, BPIC20ID, BPIC20PTC, BPIC20TPD, Sepsis, P2P), the ratio ρ_{SP} exceeds 0.7, indicating a strong alignment between predicted uncertainty and prediction error. As illustrated in Figure 1, the uncertainty-based curves closely follow the oracle and remain clearly separated from the random baseline. This behavior confirms that high-error predictions are systematically assigned higher uncertainty, and that removing uncertain predictions leads to a consistent and substantial reduction in error.

The HelpDesk event log exhibits a lower ratio ($\rho_{\mathrm{SP}} = 0.211$), yet its sparsification curve remains largely monotonic, indicating that uncertainty still provides useful, albeit weaker, guidance for identifying unreliable predictions.

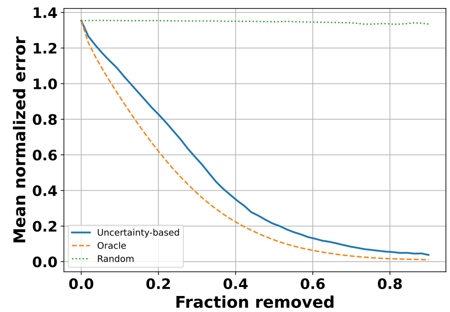
In 4 out of 14 logs (BPIC15-4, BPIC15-5, BPIC20RFP, BPIC20DD), the ratio ρ_{SP} ranges between 0.182 and 0.507, indicating moderate alignment. As shown in Figure 2, the sparsification curves deviate from monotonic behavior, particularly in the initial removal steps where the highest-uncertainty predictions are not always the most erroneous. However, beyond this initial region, the uncertainty-based curves generally lie between the oracle and random baselines and exhibit a decreasing trend. This suggests that while uncertainty estimates are noisy for the most extreme predictions, they remain informative for a substantial portion of the dataset.

Table 13: Normalized sparsification metrics evaluating the relationship between predicted uncertainty and prediction error $\rho_{\text{SP}} = \frac{nAURG}{nAUSE+nAURG}$.

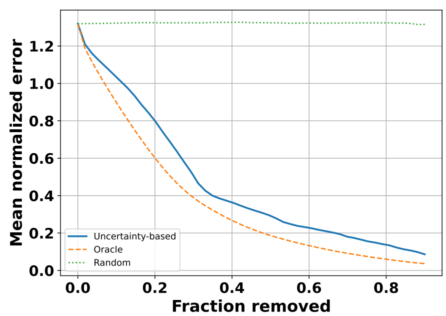
Dataset	nAUSE ↓	nAURG ↑	ρ_{SP} ↑
P2P	0.147	0.347	0.703
BPIC17W	1.077	-0.322	-0.427
BPIC15-1	0.203	0.629	0.756
BPIC15-2	1.501	-0.202	-0.156
BPIC15-3	0.095	0.817	0.896
BPIC15-4	0.486	0.108	0.182
BPIC15-5	0.308	0.317	0.507
HelpDesk	0.311	0.083	0.211
Sepsis	0.194	0.762	0.797
BPIC20ID	0.094	0.772	0.891
BPIC20DD	0.528	0.259	0.330
BPIC20PTC	0.148	0.973	0.868
BPIC20TPD	0.117	0.551	0.825
BPIC20RFP	0.430	0.289	0.402



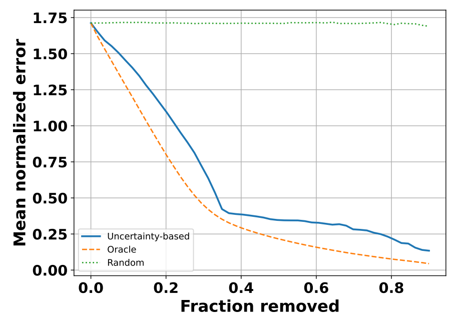
(a) BPIC15-1



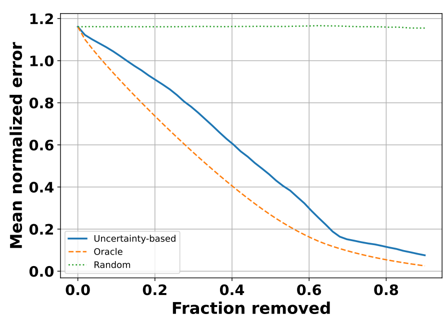
(b) BPIC15-3



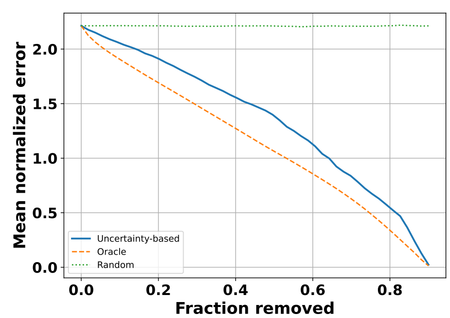
(c) BPIC20ID



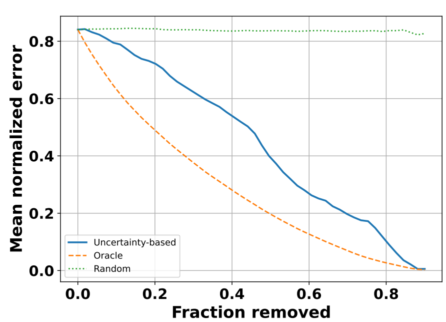
(d) BPIC20PTC



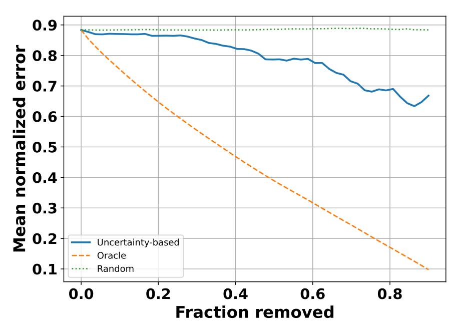
(e) BPIC20TPD



(f) Spesis

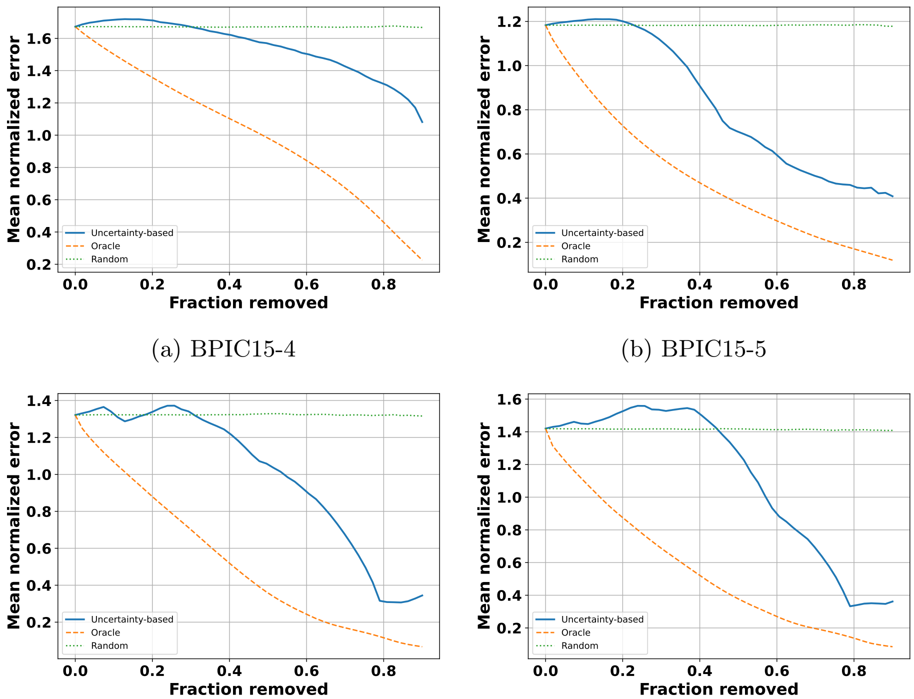


(g) P2P



(h) Helpdesk

Fig. 1: Sparsification curves with strong alignment between model's uncertainty and error.



(a) BPIC15-4 (b) BPIC15-5

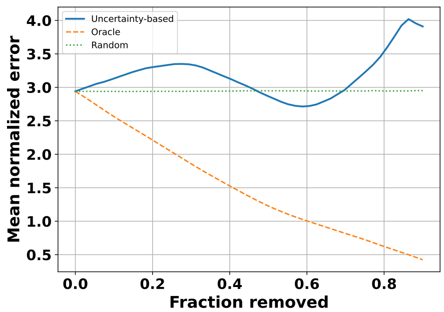
(c) BPIC20RFP (d) BPIC20DD

Fig. 2: Sparsification curves with moderate alignment and partial deviations from monotonicity.

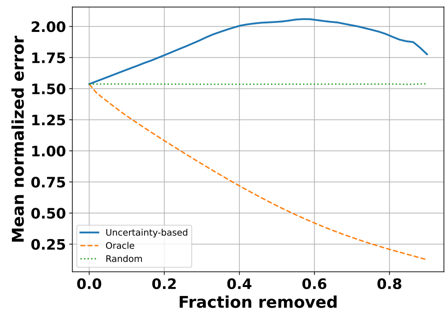
Finally, two event logs (BPIC15-2 and BPIC17W) show weak or negative alignment between uncertainty and prediction error, as reflected by low or negative values of ρ_{SP}. As shown in Figure 3, the uncertainty-based curves deviate substantially from the oracle and in some regions even approach or underperform the random baseline. This indicates that uncertainty estimates fail to reliably distinguish between high- and low-error predictions in these cases, limiting their usefulness for identifying unreliable predictions.

Uncertainty-Aware Delay Detection. We complement the main-paper results with additional evidence on how predictive uncertainty affects delay detection. Specifically, we report confusion-matrix counts for the baseline classifier and the uncertainty-aware model at the main threshold $q = 0.8$, provide an earliness analysis across different stages of case execution, and include further experiments under a stricter delay definition ($q = 0.9$). Together, these results clarify where the performance gains reported in the main paper come from and how they evolve as the definition of delay becomes more selective.

Confusion Matrix Analysis. Table 14 provides additional insight into the aggregate metrics reported in the main paper by showing the underlying confusion-



(a) BPIC15-2



(b) BPIC17W

Fig. 3: Sparsification curves with weak alignment between model's uncertainty and error.

matrix counts. The main pattern is a systematic shift from false negatives to true positives. In other words, the uncertainty-aware model identifies substantially more delayed cases than the baseline across nearly all datasets, which explains the large recall gains reported in the main paper. This improvement is particularly pronounced in logs where the baseline is overly conservative. For example, in BPIC20PTC the baseline detects only 1 delayed prefix and misses 174, whereas the uncertainty-aware model detects 132 and reduces the number of missed delayed cases to 43. Similar effects can be observed in HelpDesk (14 to 486 true positives), BPIC20DD (294 to 1257), and BPIC20TPD (91 to 575).

At the same time, the confusion matrices show that these recall gains are not obtained for free. In several datasets, the uncertainty-aware model increases the number of false positives, reflecting a more aggressive strategy for flagging potentially delayed cases. This is especially visible in HelpDesk, BPIC15-2, and BPIC20PTC, where the classifier retrieves many more delayed cases but also raises substantially more false alarms. This behavior is consistent with the precision results in the main paper: precision sometimes remains similar and sometimes decreases moderately, but the increase in true positives is often large enough to yield a clear net improvement in F1-score.

The confusion matrices also help distinguish between different types of gains. In some logs, such as BPIC17W, BPIC15-3, BPIC20DD, and BPIC20RFP, the uncertainty-aware model improves true-positive counts while keeping the growth in false positives relatively limited. In these cases, the uncertainty features appear to support a better separation between delayed and non-delayed prefixes, which explains why both recall and precision improve simultaneously. In others, such as BPIC15-4 and BPIC20ID, the gain is driven primarily by recovering delayed cases that the baseline almost always misses, even if this comes with a weaker precision trade-off. This supports the interpretation from the main paper that uncertainty is particularly valuable when the goal is not merely conservative classification, but timely retrieval of high-risk cases.

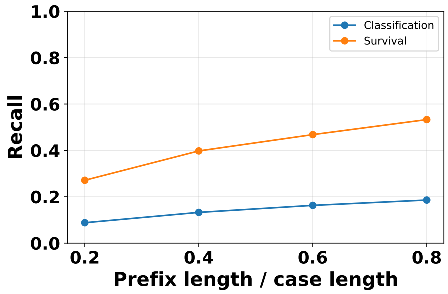
Table 14: Delay detection performance (threshold $q = 0.8$). Average confusion-matrix counts over five random seeds for baseline and uncertainty-aware models.

		Baseline Classifier				Uncertainty-Aware Model			
Dataset	# Positives	TP	TN	FP	FN	TP	TN	FP	FN
P2P	819	94	**336**	**1**	725	**684**	298	39	**135**
BPIC17W	12461	4261	21364	4683	8200	**7382**	**23779**	**2268**	**5079**
BPIC15-1	1201	591	**7717**	**1255**	610	**789**	7700	1272	**412**
BPIC15-2	359	138	**6371**	**1147**	221	**268**	5562	1956	**91**
BPIC15-3	1787	551	8516	900	1236	**1165**	**8797**	**619**	**622**
BPIC15-4	229	91	**5643**	**2357**	138	**172**	5412	2588	**57**
BPIC15-5	1629	536	**8600**	**790**	1093	**868**	8142	1248	**761**
HelpDesk	736	14	**2736**	**23**	722	**486**	1315	1444	**250**
Sepsis	519	97	**2124**	**86**	422	**266**	1524	686	**253**
BPIC20ID	251	12	**13350**	**272**	239	**99**	12879	743	**152**
BPIC20DD	2673	294	**6620**	**283**	2379	**1257**	6222	681	**1416**
BPIC20PTC	175	1	**2970**	**3**	174	**132**	2424	549	**43**
BPIC20TPD	1260	91	**11770**	**316**	1169	**575**	10616	1470	**685**
BPIC20RFP	2125	383	**3843**	**145**	1742	**1014**	3628	360	**1111**

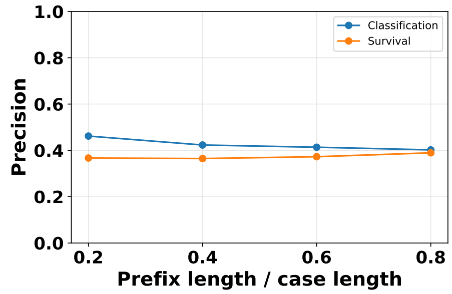
Statistical Comparison of Delay Detection Performance. To complement the aggregate performance metrics, we conduct a Wilcoxon signed-rank test to compare the baseline classifier and the uncertainty-aware model across all datasets. The results confirm that the observed improvements are statistically robust. In particular, the uncertainty-aware model achieves a significant improvement in recall ($p = 0.0002$), indicating a consistent increase in the ability to identify delayed cases. Similarly, improvements in F1-score ($p = 0.0019$) and PR-AUC ($p = 0.0007$) are statistically significant, demonstrating that the gains in recall translate into better overall performance and improved ranking of delayed cases. In contrast, the difference in precision is not statistically significant ($p = 0.27$), suggesting that the increase in recall is not accompanied by a consistent degradation in precision across datasets. This result supports the findings in the main paper: uncertainty modeling substantially improves the detection of delayed cases while maintaining competitive precision, leading to a better balance between false positives and false negatives.

Earliness Analysis. Figure 4 shows the average performance across datasets as a function of the observed fraction of each case. The plots confirm that the uncertainty-aware model is already advantageous at early stages of execution. The largest gap appears in recall: at 20% of the case length, the uncertainty-aware model achieves an average recall of roughly 0.27, compared with about 0.09 for the baseline, and this advantage persists throughout the trajectory up to 80% of the case length, where the corresponding values are about 0.53 and 0.18.

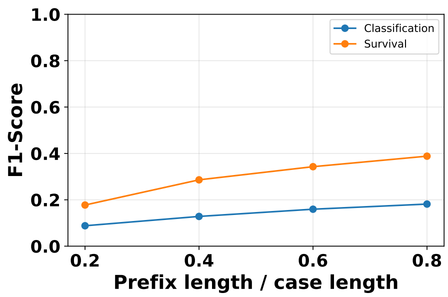
A similar pattern is visible for F1-score. The uncertainty-aware model remains above the baseline across all prefix ratios, with the largest absolute gap



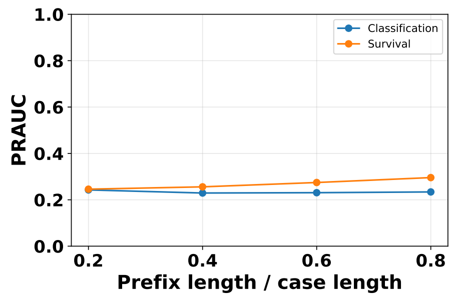
(a) Recall



(b) Precision



(c) F1-Score



(d) PRAUC

Fig. 4: Earliness performance across prefix lengths for delay detection ($q = 0.8$).

again appearing early and mid execution. Average F1-score increases from approximately 0.18 to 0.39 for the uncertainty-aware model between 20% and 80% of the case, whereas the baseline rises only from about 0.09 to 0.18. This indicates that the additional recall obtained from uncertainty modeling is not offset by a collapse in precision, but instead translates into a consistently better balance between missed and detected delayed cases.

Precision behaves differently. The baseline starts slightly higher, with average precision around 0.46 at 20% of the case length, while the uncertainty-aware model starts around 0.37. However, the gap narrows steadily as more events are observed, and by 80% the two models are nearly indistinguishable, at roughly 0.40 and 0.39, respectively. Thus, the uncertainty-aware model sacrifices little precision in later stages while maintaining substantially better recall throughout.

The PR-AUC plot provides a complementary ranking-based view. At 20% of the case length the two methods are nearly identical, both around 0.24, but the uncertainty-aware model improves steadily as more of the case is observed, reaching roughly 0.29 at 80%, while the baseline remains almost flat around 0.23–0.24. This suggests that uncertainty-derived features improve not only binary decisions at a fixed threshold, but also the overall ranking of prefixes by delay risk.

Table 15: Delay detection performance (threshold $q = 0.9$). Average Recall, Precision, F1-score, and PR-AUC over five random seeds for baseline and uncertainty-aware models.

	Baseline Classifier				Uncertainty-Aware Model			
Dataset	Recall	Precision	F1-Score	PRAUC	Recall	Precision	F1-Score	PRAUC
P2P	0.16	**1.00**	0.27	0.66	**0.85**	0.90	**0.87**	**0.85**
BPIC17W	0.19	0.29	0.23	0.20	**0.59**	**0.55**	**0.57**	**0.40**
BPIC15-1	0.31	0.07	0.11	0.04	**0.63**	**0.09**	**0.15**	**0.06**
BPIC15-2		0.00	0.00			0.00	0.00	
BPIC15-3	0.25	0.31	0.27	0.12	**0.70**	**0.50**	**0.58**	**0.37**
BPIC15-4	0.00	0.00	0.00	0.01	**0.33**	**0.01**	**0.02**	**0.01**
BPIC15-5	0.28	**0.34**	**0.30**	**0.14**	**0.52**	0.21	0.29	0.14
HelpDesk	0.00	0.11	0.00	0.11	**0.58**	**0.17**	**0.26**	**0.14**
Sepsis	0.04	0.04	0.04	0.04	**0.56**	**0.06**	**0.11**	**0.05**
BPIC20ID	0.00	0.00	0.00	0.00	**0.22**	**0.02**	**0.03**	**0.01**
BPIC20DD	0.01	0.33	0.02	0.14	**0.46**	**0.57**	**0.51**	**0.34**
BPIC20PTC	0.00		0.00	0.01	**0.72**	0.04	**0.08**	**0.03**
BPIC20TPD	0.00	0.00	0.00	0.03	**0.44**	**0.15**	**0.23**	**0.09**
BPIC20RFP	0.05	**0.83**	0.09	0.21	**0.56**	0.59	**0.57**	**0.40**

Overall, the earliness analysis supports the main-paper claim that uncertainty modeling is most useful when only limited evidence is available. The gains are strongest at early and intermediate stages, where direct classification from prefix features alone is most difficult. As the case progresses, the baseline becomes somewhat more competitive, but the uncertainty-aware model remains superior across all four metrics.

Stricter Delay Definition. Table 15 evaluates the same task under a stricter delay definition, where only the most extreme cases are labeled as delayed. This setting is more challenging because the positive class becomes rarer and the classification problem becomes more imbalanced. Nevertheless, the uncertainty-aware model preserves the same qualitative advantage observed at $q = 0.8$: it consistently achieves substantially higher recall and, in most datasets, higher F1-score and PR-AUC as well.

The stricter threshold also makes the limitations of the baseline more visible. In several logs, including BPIC15-4, HelpDesk, BPIC20ID, BPIC20PTC, and BPIC20TPD, the baseline essentially collapses to near-zero recall, indicating that it almost never retrieves the most severely delayed cases. By contrast, the uncertainty-aware model continues to detect a non-trivial fraction of these rare events. This is particularly important from an operational perspective, since the cases defined as delayed at $q = 0.9$ correspond to the extreme tail where early intervention is likely to be most valuable.

At the same time, the table shows that the harder decision problem amplifies the precision–recall trade-off. Because positives are rarer, any increase in recall is more easily accompanied by a reduction in precision. This can be seen in datasets

Table 16: Delay detection performance (threshold $q = 0.9$). Average confusion-matrix counts over five random seeds for baseline and uncertainty-aware models.

		Baseline Classifier				Uncertainty-Aware Model			
Dataset	# Positives	TP	TN	FP	FN	TP	TN	FP	FN
P2P	693	111	**463**	**0**	582	**589**	396	67	**104**
BPIC17W	6642	1270	**28767**	**3099**	5372	**3938**	28673	3193	**2704**
BPIC15-1	173	54	**9228**	**772**	119	**109**	8852	1148	**64**
BPIC15-2	0	0	**7619**	**258**	0	0	7329	548	0
BPIC15-3	669	165	**10167**	**367**	504	**469**	10064	470	**200**
BPIC15-4	45	0	**7469**	**715**	45	**15**	6430	1754	**30**
BPIC15-5	620	175	**10057**	**342**	445	**325**	9068	1331	**295**
HelpDesk	387	0	**3107**	**1**	387	**224**	1980	1128	**163**
Sepsis	109	5	**2519**	**101**	104	**61**	1627	993	**48**
BPIC20ID	32	0	**13737**	**104**	32	**7**	13445	396	**25**
BPIC20DD	1353	13	**8196**	**27**	1340	**623**	7750	473	**730**
BPIC20PTC	17	0	**3131**	**0**	17	**12**	2850	281	**5**
BPIC20TPD	452	0	**12818**	**76**	452	**201**	11786	1108	**251**
BPIC20RFP	1065	53	**5037**	**11**	1012	**597**	4624	424	**468**

such as BPIC15-1, Sepsis, and BPIC20PTC. Even in these cases, however, the uncertainty-aware model usually yields better F1-score and PR-AUC, indicating that the additional retrieved positives outweigh the extra false alarms. In other logs, such as P2P, BPIC15-3, BPIC20DD, and BPIC20RFP, the gains are strong across multiple metrics, suggesting that uncertainty remains informative even under a more selective delay definition.

The two excluded cases should be interpreted separately. For BPIC15-2, the test set contains no delayed cases at $q = 0.9$, so recall-based metrics are undefined. For BPIC20PTC, the positive class is extremely small, which makes the estimates inherently unstable. These edge cases illustrate an important practical point: as the delay threshold becomes more extreme, the evaluation becomes increasingly sensitive to the absolute number of positive examples. Even under these conditions, however, the broader pattern remains unchanged: uncertainty modeling improves the retrieval of high-delay cases.

Table 15 summarizes these results, confirming that the benefits of uncertainty modeling persist under more stringent delay thresholds and further supporting its practical utility for prioritizing delayed cases.

The confusion matrices in Table 16 clarify how the stricter threshold changes the prediction behavior. Relative to $q = 0.8$, the positive class is much smaller, so both models face a harder retrieval problem. The baseline responds by becoming extremely conservative: in many datasets it produces either no true positives at all or only a handful, which explains the near-zero recall values in Table 15. Examples include BPIC20ID, BPIC20PTC, BPIC20TPD (zero true positives for all of them).

The uncertainty-aware model remains much more effective at recovering rare delayed cases. For instance, it increases the number of true positives from 13 to 623 in BPIC20DD, from 53 to 597 in BPIC20RFP, and from 0 to 201 in BPIC20TPD. Even where precision is modest, these gains are practically important because they correspond to cases from the extreme tail of the cycle-time distribution. In other words, the uncertainty-aware model continues to identify instances that the baseline classifier systematically fails to retrieve.

As expected, this increased sensitivity can come with larger false-positive counts. This is particularly visible in logs such as BPIC20TPD, HelpDesk, and Sepsis. However, the confusion matrices show that the uncertainty-aware model is not simply predicting the positive class indiscriminately. Rather, it shifts the operating point toward a substantially better recall regime, while still preserving useful discrimination in several datasets. This is consistent with the PR-AUC improvements reported in Table 15, which indicate that uncertainty-derived features improve the ranking of prefixes by delay risk, even when a single decision threshold may still involve a precision–recall trade-off.

Statistical Comparison under a Stricter Delay Threshold. We further assess statistical significance under the stricter delay definition ($q = 0.9$) using the Wilcoxon signed-rank test. The results confirm that the benefits of uncertainty modeling remain consistent even when focusing on the most extreme delayed cases. In particular, the uncertainty-aware model achieves a statistically significant improvement in recall ($p < 0.001$), indicating that it consistently identifies a larger proportion of severely delayed cases compared to the baseline. Improvements in F1-score ($p = 0.002$) and PR-AUC ($p < 0.001$) are also statistically significant, demonstrating that the increased recall translates into better overall performance and improved ranking of high-risk cases. In contrast, the difference in precision is not statistically significant ($p = 0.27$), reflecting the stronger class imbalance and the inherent difficulty of maintaining high precision when detecting rare extreme delays.

Overall, these findings reinforce the conclusions drawn for $q = 0.8$: uncertainty modeling provides a statistically robust improvement in identifying delayed cases, and its benefits persist even under more stringent delay definitions where the task becomes substantially more challenging.

Summary. Taken together, the additional results strengthen the conclusions of the main paper. The confusion matrices show that the improvements in recall are driven by a substantial reduction in missed delayed cases rather than by isolated dataset-specific effects. The earliness analysis demonstrates that these gains emerge early in the execution of a case, when timely intervention is most valuable. Finally, the experiments at $q = 0.9$ show that the benefits of uncertainty modeling persist even when delay detection focuses only on the most extreme cases. These results further support the view that predictive uncertainty provides actionable information beyond prefix features alone, enabling earlier and more reliable prioritization of high-risk process instances.

References

1. Branco, P., Ribeiro, R.P., Torgo, L.: Ubl: an r package for utility-based learning. arXiv preprint arXiv:1604.08079 (2016)
2. Ren, J., Zhang, M., Yu, C., Liu, Z.: Balanced mse for imbalanced visual regression. In: CVPR. pp. 7926–7935 (2022)
3. Ribeiro, R.P., Moniz, N.: Imbalanced regression and extreme value prediction. Machine Learning **109**(9), 1803–1835 (2020)
4. Silva, A., Ribeiro, R.P., Moniz, N.: Model optimization in imbalanced regression. In: International Conference on Discovery Science. pp. 3–21. Springer (2022)
5. Torgo, L., Ribeiro, R.: Utility-based regression. In: European conference on principles of data mining and knowledge discovery. pp. 597–604. Springer (2007)
6. Yang, Y., Zha, K., Chen, Y., Wang, H., Katabi, D.: Delving into deep imbalanced regression. In: ICML. pp. 11842–11851. PMLR (2021)